



KeraNova

Explainable Prediction of 12-Month Visual Outcomes

Project Idea

THE CLINICAL NEED

- Visual and refractive recovery after keratoconus treatment varies between eyes.
- Published comparisons primarily describe average outcomes at the group level.
- Preoperative counselling would benefit from transparent, patient-level outcome estimates.

THE AI-ASSISTED APPROACH

- Use routinely collected baseline clinical and Pentacam variables from the retrospective cohort.
- Estimate 12-month corrected distance visual acuity after either procedure.
- Develop an interpretable research prototype—not an autonomous treatment recommendation.

Primary research question:

Can baseline clinical and tomographic characteristics predict 12-month CDVA after TG-PRK + accelerated CXL versus accelerated CXL alone?

Objectives and Aims

1

Predict 12-month CDVA

Develop a parsimonious model using baseline CDVA, treatment group and selected clinical and tomographic variables.

2

Quantify predictive performance

Report prediction error, calibration and uncertainty using patient-level internal validation.

3

Identify informative predictors

Describe the direction, magnitude and uncertainty of the strongest predictive associations.

4

Define responsible clinical scope

Produce an exploratory research prototype to support future counselling research—not clinical decision support.

Methods & Required Data

1

Dataset and cohort

Authorized de-identified EHR and Pentacam export; up to 60 eyes (30 per group), KKESH, 2020–2025.

2

Primary outcome

Twelve-month CDVA in logMAR within the protocol-defined 10–14-month follow-up window.

3

Candidate predictors

Baseline CDVA, age, sex, treatment group, Kmax and one prespecified pachymetry metric.

4

Data quality

Eligibility audit, unit harmonization, missingness assessment and identification of bilateral-eye data.

5

Modeling

Penalized linear regression benchmarked against a prespecified clinical baseline model.

6

Internal validation

Patient-level resampling; MAE, RMSE, calibration and 95% confidence intervals.

Three-Day Execution Plan

1

Day 1 | Curate

Receive the authorized export; confirm eligibility, completeness and absence of direct identifiers.

2

Day 1 | Define

Freeze the outcome, predictor set and analysis plan with cornea and biostatistics leads.

3

Day 2 | Analyze

Run the cohort comparison and fit the prespecified transparent prediction model.

4

Day 2 | Validate

Perform patient-level resampling, calibration assessment and sensitivity analyses.

5

Day 3 | Interpret

Review clinical plausibility, uncertainty and failure modes with a clinician in the loop.

6

Day 3 | Deliver

Produce reproducible tables, figures and a manuscript-ready results section.

KKESH Expertise and Governance

Required KKESH clinical expertise

- Cornea and refractive surgery: confirm eligibility, endpoint definition and clinical interpretation.
- Biostatistics: finalize model complexity, patient-level resampling and uncertainty reporting.
- Research data and Pentacam support: authorize extraction, maintain the data dictionary and verify measurement quality.

IRB and data governance

The parent retrospective study has received KKESH IRB approval. The AI extension will undergo the required KKESH IRB amendment or formal determination before execution. The .analytic file will use coded patient IDs only, exclude MRN and remain within authorized KKESH systems

Core Team and Availability

Name	Position	Hackathon contribution	Availability: 1–3 Oct 2026
Dr. Ahmed Aldayel	Consultant, Cornea & Refractive Surgery	Clinical lead; endpoint adjudication	Yes
Ahmed Hejazi	Software Engineering & Artificial Intelligence	AI/model development; software implementation	Yes
Mustafa Taher	Software Engineering & Artificial Intelligence	AI/model development; software implementation	Yes
Raghad Saleh Alajlan	Medical Intern	Clinical data QA; manuscript drafting	Yes
Abdulrhman Al-Ammar	Medical Student	Analysis support; presentation	yes
Renad Ali Al-Shehri	Medical Student	Data curation; literature review	yes